Week 6: Bulletproof Intake

ITSS/OPRE 3312 · `intake_safe.py` + `test_intake.py` · Data: `data/wk06_data_raw_cases_100.csv`

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Rows loaded cleanly	Rows quarantined	pytest tests passed

The goal

Load 100 raw case records, keep every row that can be loaded, set aside the damaged ones with a reason, and never crash. The two piles must add up to 100.

The three traps in the data

1. Truncated rows 6 rows

Some rows have 6 fields instead of 8. `parse_row` checks `len(row) != 8` first and raises a `ValueError`, so reaching for `row[7]` never causes an `IndexError`.

2. Age is `'unknown'` 3 rows

`int(row[3])` raises `ValueError` by itself. No extra check needed. It's caught in `load_cases`.

3. Empty date 2 rows

An empty string doesn't crash anything, so the code checks `row[4].strip() == ""` itself and raises `ValueError("missing date")`.

How it works

`parse_row(row)`

One CSV row in, one clean dictionary out. It strips spaces, title-cases names and addresses, upper-cases sex and status, and converts age to an `int`. If the row is damaged it **raises** instead of returning.

`load_cases(path)`

Reads the file with `csv.reader`, which keeps quoted names like `"KIRKLAND, MONICA, JR, III"` in one piece. It skips the header with `next()` and blank lines with `continue`, and wraps each `parse_row` call in `try / except ValueError`. It returns two lists: `good` and `quarantine`.

```
try:
    good.append(parse_row(row))
except ValueError as err:
    quarantine.append((line_num, str(err)))
```

`if __name__ == "__main__":`

The report only prints when the file is run directly. When `test_intake.py` imports `parse_row`, the report is skipped.

Quarantine log

Line	Reason
2	missing date
17	invalid literal for int() with base 10: 'unknown'
26	expected 8 fields, got 6
30	expected 8 fields, got 6
34	expected 8 fields, got 6
42	missing date
43	expected 8 fields, got 6
44	expected 8 fields, got 6
50	invalid literal for int() with base 10: 'unknown'
64	invalid literal for int() with base 10: 'unknown'
86	expected 8 fields, got 6

Tests (`pytest test_intake.py`)

Test	What it proves	Result
`test_good_row_parses`	Age `"36"` becomes the number 36, and `" carter, d "` becomes `"Carter, D"`	passed
`test_bad_age_rejected`	Age `'unknown'` raises `ValueError`	passed
`test_missing_date_rejected`	An empty date raises `ValueError`, and only the date is wrong	passed

```
$ pytest -q test_intake.py
...                                                                      [100%]
3 passed in 0.01s
```